

CSA43 – INTERNET PROGRAMMING
Assessment Tool 2 – Pseudocode Writing Task

QUESTION 1 – XML Parsing Algorithm

Write a detailed pseudocode algorithm to parse an XML document that contains structured data such as user or product details. Your algorithm should clearly describe the steps involved in loading the XML file, accessing its elements, traversing through nodes, extracting required data, and displaying the output in a readable format. Ensure that the logic handles hierarchical structure properly.

What the algorithm must cover

Key idea: Load → Parse → Root → Parent nodes → Traverse child nodes → Extract values → Display.

Pseudocode Answer

```
ALGORITHM XML_PARSING
INPUT: XML file containing structured user/product data
OUTPUT: Required XML data displayed in readable format

BEGIN
    1. Specify the XML file path.
    2. Check whether the XML file exists.
        IF file does not exist THEN
            Display "XML file not found"
            STOP
        END IF

    3. Load the XML document.
        IF loading fails THEN
            Display "Unable to load XML document"
            STOP
        END IF

    4. Parse the XML document.

    5. Access the root node of the XML document.

    6. Identify the required parent elements
        such as USERS or PRODUCTS.

    7. Traverse each child node under the parent element.
        FOR each record node DO

            7.1 Access the required child elements.
            7.2 Extract values such as ID, NAME, EMAIL,
                PRODUCT_NAME, PRICE, etc.
            7.3 Check whether a required value is available.
            7.4 Store or prepare the extracted values.

        END FOR

    8. If a record contains nested child elements,
        traverse the child elements in the same manner.

    9. Format the extracted information into a
        readable output.

    10. Display the extracted records.

    11. Display "XML parsing completed".

END
```

QUESTION 2 – MVC Workflow

Write a pseudocode representation of the MVC (Model–View–Controller) workflow in a web application. Your answer should clearly illustrate how a user request is received, processed by the Controller, how the Model interacts with the database, and how the View generates and returns the response to the user. Include all major steps involved in request handling and response generation.

What the algorithm must cover

Key idea: User → Controller → Model → Database → Model → Controller → View → User.

Pseudocode Answer

```
ALGORITHM MVC_WORKFLOW
INPUT: User request
OUTPUT: Response displayed to the user

BEGIN
    1. User sends a request from the browser.

    2. Web application receives the request.

    3. Controller identifies the requested operation.

    4. Controller validates the request information.

    5. Controller sends the required operation
       to the Model.

    6. Model connects to the database.

       IF database connection fails THEN
           Return an error status/message to Controller
           Controller selects an error response
           View displays the error
           STOP
       END IF

    7. Model executes the required database operation.

    8. Model retrieves/processes the required data.

    9. Model sends the result back to Controller.

    10. Controller selects the appropriate View.

    11. Controller passes the processed data to View.

    12. View prepares the response page.

    13. View generates the response.

    14. Server sends the response to the browser.

    15. Browser displays the result to the user.

END
```

QUESTION 3 – JSP Request Flow

Write a detailed pseudocode algorithm to explain the request–response flow in a JSP-based web application. Your algorithm should include steps such as client request initiation, server processing, interaction with backend components (like Servlets or database), and generation of dynamic content that is sent back to the client.

What the algorithm must cover

Key idea: Browser request → Server/JSP → Servlet/backend → Database if needed → JSP generates content → Browser.

Pseudocode Answer

```
ALGORITHM JSP_REQUEST_FLOW
INPUT: Client request
OUTPUT: Dynamic web response
```

```
BEGIN
```

1. User opens the web page or submits a request.
2. Browser sends the request to the web server.
3. Server receives the JSP request.
4. Identify the requested JSP page.
5. Process the JSP page on the server.
6. If a Servlet/Controller is involved,
 forward the request to the appropriate component.
7. Servlet/Controller processes the request.
8. If backend data is required THEN
 Establish database connection.
 Execute the required data operation.
 Retrieve the required result.
 Close the database connection.
 END IF
9. Pass the required data to the JSP page.
10. JSP combines the page structure with
 the dynamic data.
11. Generate the final response content.
12. Server sends the generated response
 to the browser.
13. Browser receives and displays the
 dynamic web page.

```
END
```

QUESTION 4 – Database Retrieval

Write a pseudocode algorithm to retrieve data from a database. Your answer should include steps for establishing a database connection, executing SQL queries, processing the result set, and displaying the retrieved data. Clearly represent the sequence of operations involved in database interaction.

What the algorithm must cover

Key idea: Connect → Prepare SELECT → Execute → Result Set → Loop through records → Display → Close.

Pseudocode Answer

```
ALGORITHM DATABASE_RETRIEVAL
INPUT: Database details and retrieval query
OUTPUT: Retrieved records displayed to the user

BEGIN
    1. Define database connection details.

    2. Load/initialize the database connectivity mechanism.

    3. Establish a connection to the database.

        IF connection fails THEN
            Display "Database connection failed"
            STOP
        END IF

    4. Prepare the required SQL SELECT query.

    5. Execute the SQL query.

        IF query execution fails THEN
            Display "Query execution failed"
            Close database connection
            STOP
        END IF

    6. Receive the result set.

    7. Check whether the result set contains records.

        IF no records are found THEN
            Display "No records found"
        ELSE
            FOR each record in the result set DO
                Read the required fields.
                Store or format the values.
                Display the record.
            END FOR
        END IF

    8. Close the result set if required.

    9. Close the database connection.

    10. Display "Data retrieval completed".

END
```

QUESTION 5 – PHP Validation

Write a pseudocode algorithm for validating user input in a web form using PHP concepts. The algorithm should include checks for empty fields, validation of input formats (such as email and password), handling invalid inputs with appropriate error messages, and allowing submission only when all validations are satisfied.

What the algorithm must cover

Key idea: Receive → Clean → Check empty → Validate format → Collect errors → If errors, reject; otherwise submit.

Pseudocode Answer

```
ALGORITHM PHP_FORM_VALIDATION
INPUT: User form containing name, email, and password
OUTPUT: Error messages OR successful form submission

BEGIN
    1. Receive form data from the user.

    2. Initialize an empty error list.

    3. Read the name, email, and password fields.

    4. Remove unnecessary spaces from the input values.

    5. Check whether the name is empty.
        IF name is empty THEN
            Add "Name is required" to error list.
        END IF

    6. Check whether the email is empty.
        IF email is empty THEN
            Add "Email is required" to error list.
        ELSE
            Check whether email has a valid email format.
            IF email format is invalid THEN
                Add "Enter a valid email address" to error list.
            END IF
        END IF

    7. Check whether the password is empty.
        IF password is empty THEN
            Add "Password is required" to error list.
        ELSE
            Check whether password satisfies the
            required length/format.
            IF password is invalid THEN
                Add "Password does not meet requirements"
                to error list.
            END IF
        END IF

    8. Check the error list.

        IF error list is not empty THEN
            Display all appropriate error messages.
            Do not submit the form.
        ELSE
            Accept the valid form data.
            Continue with form submission/processing.
            Display "Form submitted successfully".
        END IF
```

END

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